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Straight Lines and Pair of Straight Lines

JEE Main Mathematics · Previous Year Questions with Answer Key

206	152	2002 – 2026	17	English
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MCQ (Single Correct Answer) 166 questions

Q1. JEE Main 2026 (Online) 8th April Evening Shift

If a straight line drawn through the point of intersection of the lines $4x + 3y - 1 = 0$ and $3x + 4y - 1 = 0$, meets the co-ordinate axes at the points P and Q , then the locus of the mid point of PQ is :

(A) $x + y - 7 = 0$ (B) $x + y - 14xy = 0$ (C) $2x + y + 14xy = 0$ (D) $x + 2y - 14xy = 0$

Open solution

Q2. JEE Main 2026 (Online) 5th April Morning Shift

In an equilateral triangle PQR, let the vertex P be at $(3, 5)$ and the side QR be along the line $x + y = 4$. If the orthocentre of the triangle PQR is (α, β), then $9(\alpha + \beta)$ is equal to:

(A) 16 (B) 27 (C) 36 (D) 48

Open solution

Q3. JEE Main 2026 (Online) 4th April Morning Shift

Let the line $L_1 : x + 3 = 0$ intersect the lines $L_2 : x - y = 0$ and $L_3 : 3x + y = 0$ at the points A and B , respectively. Let the bisector of the obtuse angle between the lines L_2 and L_3 intersect the line L_1 at the point C. Then $BC^2 : AC^2$ is equal to:

(A) 5:1 (B) 1:5 (C) 2:3 (D) 3:2

Open solution

Q4. JEE Main 2026 (Online) 4th April Morning Shift

Let the vertex A of a triangle ABC be $(1, 2)$, and the mid-point of the side AB be $(5, -1)$. If the centroid of this triangle is $(3, 4)$ and its circumcenter is (α, β), then $21(\alpha + \beta)$ is equal to :

(A) 309 (B) 403 (C) 497 (D) 524

Open solution

Q5. JEE Main 2026 (Online) 2nd April Morning Shift

Let the mid points of the sides of a triangle ABC be $\left(\frac{5}{2}, 7\right)$, $\left(\frac{5}{2}, 3\right)$ and $(4, 5)$. If its incentre is (h, k), then $3h + k$ is equal to :

(A) 11 (B) 12 (C) 13 (D) 14

Open solution

Q6. JEE Main 2026 (Online) 28th January Morning Shift

Let ABC be an equilateral triangle with orthocenter at the origin and the side BC on the line $x + 2\sqrt{2}y = 4$. If the co-ordinates of the vertex A are (α, β), then the greatest integer less than or equal to $|\alpha + \sqrt{2}\beta|$ is

(A) 5 (B) 4 (C) 2 (D) 3

Open solution

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Q7. JEE Main 2026 (Online) 24th January Evening Shift

Let the angles made with the positive x-axis by two straight lines drawn from the point $\mathrm{P}(2,3)$ and meeting the line $x+y=6$ at a distance $\sqrt{\frac{2}{3}}$ from the point P be θ_1 and θ_2. Then the value of $(\theta_1+\theta_2)$ is:

Open solution

(A) $\frac{\pi}{2}$ (B) $\frac{\pi}{3}$ (C) $\frac{\pi}{12}$ (D) $\frac{\pi}{6}$

Q8. JEE Main 2026 (Online) 24th January Morning Shift

Let $A(1,0), B(2,-1)$ and $C\left(\frac{7}{3}, \frac{4}{3}\right)$ be three points. If the equation of the bisector of the angle ABC is $\alpha x+\beta y=5$, then the value of $\alpha^2+\beta^2$ is

Open solution

(A) 5 (B) 10 (C) 8 (D) 13

Q9. JEE Main 2026 (Online) 23rd January Evening Shift

Let $\mathrm{A}(1,2)$ and $\mathrm{C}(-3,-6)$ be two diagonally opposite vertices of a rhombus, whose sides AD and BC are parallel to the line $7x-y=14$. If $\mathrm{B}(\alpha,\beta)$ and $\mathrm{D}(\gamma,\delta)$ are the other two vertices, then $|\alpha+\beta+\gamma+\delta|$ is equal to :

Open solution

(A) 3 (B) 6 (C) 1 (D) 9

Q10. JEE Main 2026 (Online) 23rd January Morning Shift

A rectangle is formed by the lines $x=0, y=0, x=3$ and $y=4$. Let the line L be perpendicular to $3x+y+6=0$ and divide the area of the rectangle into two equal parts. Then the distance of the point $\left(\frac{1}{2},-5\right)$ from the line L is equal to :

Open solution

(A) $\sqrt{10}$ (B) $2\sqrt{5}$ (C) $2\sqrt{10}$ (D) $3\sqrt{10}$

Q11. JEE Main 2026 (Online) 22nd January Evening Shift

Among the statements

$(S1)$: If $A(5,-1)$ and $B(-2,3)$ are two vertices of a triangle, whose orthocentre is $(0,0)$, then its third vertex is $(-4,-7)$

and

(S2) : If positive numbers $2a, b, c$ are three consecutive terms of an A.P., then the lines $ax+by+c=0$ are concurrent at $(2,-2)$,

Open solution

(A) both are incorrect (B) only (S2) is correct

(C) both are correct (D) only (S1) is correct

Q12. JEE Main 2026 (Online) 21st January Morning Shift

Let a point A lie between the parallel lines L_1 and L_2 such that its distances from L_1 and L_2 are 6 and 3 units, respectively. Then the area (in sq. units) of the equilateral triangle ABC , where the points B and C lie on the lines L_1 and L_2, respectively, is :

Open solution

(A) $21\sqrt{3}$ (B) $12\sqrt{2}$ (C) $15\sqrt{6}$ (D) 27

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Q13. JEE Main 2025 (Online) 8th April Evening Shift

Let a be the length of a side of a square OABC with O being the origin. Its side OA makes an acute angle α with the positive x-axis and the equations of its diagonals are $(\sqrt{3}+1)x+(\sqrt{3}-1)y=0$ and $(\sqrt{3}-1)x-(\sqrt{3}+1)y+8\sqrt{3}=0$. Then a^2 is equal to :

Open solution

(A) 48 (B) 16 (C) 24 (D) 32

Q14. JEE Main 2025 (Online) 8th April Evening Shift

A line passing through the point *P(a, 0)* makes an acute angle α with the positive x-axis. Let this line be rotated about the point *P* through an angle $\frac{\alpha}{2}$ in the clockwise direction. If in the new position, the slope of the line is $2-\sqrt{3}$ and its distance from the origin is $\frac{1}{\sqrt{2}}$, then the value of $3a^2\tan^2\alpha-2\sqrt{3}$ is :

Open solution

(A) 8 (B) 4 (C) 5 (D) 6

Q15. JEE Main 2025 (Online) 7th April Evening Shift

If the orthocenter of the triangle formed by the lines y = x + 1, y = 4x - 8 and y = mx + c is at (3, -1), then m - c is :

Open solution

(A) 0 (B) 2 (C) -2 (D) 4

Q16. JEE Main 2025 (Online) 7th April Morning Shift

Let ABC be the triangle such that the equations of lines AB and AC be $3y-x=2$ and $x+y=2$, respectively, and the points B and C lie on x-axis. If P is the orthocentre of the triangle ABC , then the area of the triangle PBC is equal to

Open solution

(A) 8 (B) 4 (C) 10 (D) 6

Q17. JEE Main 2025 (Online) 4th April Morning Shift

Let the three sides of a triangle lie on the lines $4x-7y+10=0, x+y=5$ and $7x+4y=15$. Then the distance of its orthocentre from the orthocentre of the triangle formed by the lines $x=0, y=0$ and $x+y=1$ is

Open solution

(A) $\sqrt{20}$ (B) 20 (C) $\sqrt{5}$ (D) 5

Q18. JEE Main 2025 (Online) 3rd April Evening Shift

Consider the lines $x(3\lambda+1)+y(7\lambda+2)=17\lambda+5, \lambda$ being a parameter, all passing through a point P. One of these lines (say L) is farthest from the origin. If the distance of L from the point $(3,6)$ is d, then the value of d^2 is

Open solution

(A) 10 (B) 20 (C) 15 (D) 30

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Q19. JEE Main 2025 (Online) 3rd April Morning Shift

A line passes through the origin and makes equal angles with the positive coordinate axes. It intersects the lines $L_1 : 2x + y + 6 = 0$ and $L_2 : 4x + 2y - p = 0, p > 0$, at the points A and B , respectively. If $AB = \frac{9}{\sqrt{2}}$ and the foot of the perpendicular from the point A on the line L_2 is M, then $\frac{AM}{BM}$ is equal to

(A) 5 (B) 3 (C) 2 (D) 4

Open solution

Q20. JEE Main 2025 (Online) 2nd April Evening Shift

Let the area of the triangle formed by a straight line $L : x + by + c = 0$ with co-ordinate axes be 48 square units. If the perpendicular drawn from the origin to the line L makes an angle of 45° with the positive x-axis, then the value of $b^2 + c^2$ is :

(A) 90 (B) 83 (C) 93 (D) 97

Open solution

Q21. JEE Main 2025 (Online) 29th January Evening Shift

Let the line $x + y = 1$ meet the axes of x and y at A and B, respectively. A right angled triangle AMN is inscribed in the triangle OAB, where O is the origin and the points M and N lie on the lines OB and AB, respectively. If the area of the triangle AMN is $\frac{4}{9}$ of the area of the triangle OAB and $AN : NB = \lambda : 1$, then the sum of all possible value(s) of λ is:

(A) $\frac{1}{2}$ (B) $\frac{5}{2}$ (C) 2 (D) $\frac{13}{6}$

Open solution

Q22. JEE Main 2025 (Online) 29th January Morning Shift

Let ΔABC be a triangle formed by the lines 7x − 6y + 3 = 0, x + 2y − 31 = 0 and 9x − 2y − 19 = 0. Let the point (h, k) be the image of the centroid of ΔABC in the line 3x + 6y − 53 = 0. Then $h^2 + k^2 + hk$ is equal to :

(A) 47 (B) 37 (C) 40 (D) 36

Open solution

Q23. JEE Main 2025 (Online) 28th January Evening Shift

Two equal sides of an isosceles triangle are along $-x + 2y = 4$ and $x + y = 4$. If m is the slope of its third side, then the sum, of all possible distinct values of m, is:

(A) $-2\sqrt{10}$ (B) 12 (C) -6 (D) 6

Open solution

Q24. JEE Main 2025 (Online) 28th January Evening Shift

If A and B are the points of intersection of the circle $x^2 + y^2 - 8x = 0$ and the hyperbola $\frac{x^2}{9} - \frac{y^2}{4} = 1$ and a point P moves on the line $2x - 3y + 4 = 0$, then the centroid of ΔPAB lies on the line :

(A) $x + 9y = 36$ (B) $9x - 9y = 32$ (C) $4x - 9y = 12$ (D) $6x - 9y = 20$

Open solution

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Q25. JEE Main 2025 (Online) 24th January Evening Shift

Let the points $\left(\frac{11}{2}, \alpha\right)$ lie on or inside the triangle with sides $x + y = 11, x + 2y = 16$ and $2x + 3y = 29$. Then the product of the smallest and the largest values of α is equal to :

(A) 22 (B) 33 (C) 55 (D) 44

Open solution

Q26. JEE Main 2025 (Online) 24th January Morning Shift

Let the lines $3x - 4y - \alpha = 0, 8x - 11y - 33 = 0$, and $2x - 3y + \lambda = 0$ be concurrent. If the image of the point $(1, 2)$ in the line $2x - 3y + \lambda = 0$ is $\left(\frac{57}{13}, \frac{-40}{13}\right)$, then $|\alpha\lambda|$ is equal to

(A) 91 (B) 113 (C) 101 (D) 84

Open solution

Q27. JEE Main 2025 (Online) 23rd January Evening Shift

A rod of length eight units moves such that its ends A and B always lie on the lines $x - y + 2 = 0$ and $y + 2 = 0$, respectively. If the locus of the point P, that divides the rod AB internally in the ratio $2 : 1$ is $9\left(x^2 + \alpha y^2 + \beta xy + \gamma x + 28y\right) - 76 = 0$, then $\alpha - \beta - \gamma$ is equal to :

(A) 24 (B) 22 (C) 21 (D) 23

Open solution

Q28. JEE Main 2025 (Online) 23rd January Morning Shift

Let the area of a $\triangle PQR$ with vertices $P(5, 4), Q(-2, 4)$ and $R(a, b)$ be 35 square units. If its orthocenter and centroid are $O\left(2, \frac{14}{5}\right)$ and $C(c, d)$ respectively, then $c + 2d$ is equal to

(A) 3 (B) $\frac{7}{3}$ (C) 2 (D) $\frac{8}{3}$

Open solution

Q29. JEE Main 2025 (Online) 22nd January Morning Shift

Let the triangle PQR be the image of the triangle with vertices $(1, 3), (3, 1)$ and $(2, 4)$ in the line $x + 2y = 2$. If the centroid of $\triangle$PQR is the point (α, β), then $15(\alpha - \beta)$ is equal to :

(A) 21 (B) 19 (C) 22 (D) 24

Open solution

Q30. JEE Main 2024 (Online) 9th April Morning Shift

A variable line L passes through the point $(3, 5)$ and intersects the positive coordinate axes at the points A and B. The minimum area of the triangle OAB, where O is the origin, is :

(A) 35 (B) 25 (C) 30 (D) 40

Open solution

Q31. JEE Main 2024 (Online) 9th April Morning Shift

A ray of light coming from the point $\mathrm{P}(1, 2)$ gets reflected from the point Q on the x-axis and then passes through the point $R(4, 3)$. If the point $S(h, k)$ is such that $PQRS$ is a parallelogram, then hk^2 is equal to:

(A) 60 (B) 70 (C) 80 (D) 90

Open solution

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Q32. JEE Main 2024 (Online) 8th April Evening Shift

If the line segment joining the points $(5, 2)$ and $(2, a)$ subtends an angle $\frac{\pi}{4}$ at the origin, then the absolute value of the product of all possible values of a is :

(A) 4 (B) 8 (C) 6 (D) 2

Open solution

Q33. JEE Main 2024 (Online) 8th April Morning Shift

The equations of two sides AB and AC of a triangle ABC are $4x + y = 14$ and $3x - 2y = 5$, respectively. The point $\left(2, -\frac{4}{3}\right)$ divides the third side BC internally in the ratio $2 : 1$, the equation of the side BC is

(A) $x + 6y + 6 = 0$ (B) $x - 3y - 6 = 0$ (C) $x + 3y + 2 = 0$ (D) $x - 6y - 10 = 0$

Open solution

Q34. JEE Main 2024 (Online) 6th April Evening Shift

If the locus of the point, whose distances from the point $(2, 1)$ and $(1, 3)$ are in the ratio $5 : 4$, is $ax^2 + by^2 + cxy + dx + ey + 170 = 0$, then the value of $a^2 + 2b + 3c + 4d + e$ is equal to :

(A) 37 (B) -27 (C) 437 (D) 5

Open solution

Q35. JEE Main 2024 (Online) 6th April Morning Shift

Let a variable line of slope $m > 0$ passing through the point $(4, -9)$ intersect the coordinate axes at the points A and B. The minimum value of the sum of the distances of A and B from the origin is

(A) 30 (B) 15 (C) 10 (D) 25

Open solution

Q36. JEE Main 2024 (Online) 5th April Evening Shift

Let $\mathrm{A}(-1, 1)$ and $\mathrm{B}(2, 3)$ be two points and P be a variable point above the line AB such that the area of $\triangle\mathrm{PAB}$ is 10. If the locus of P is $\mathrm{a}x + \mathrm{by} = 15$, then $5\mathrm{a} + 2\,\mathrm{b}$ is :

(A) $-\frac{12}{5}$ (B) $-\frac{6}{5}$ (C) 6 (D) 4

Open solution

Q37. JEE Main 2024 (Online) 5th April Morning Shift

Let two straight lines drawn from the origin O intersect the line $3x + 4y = 12$ at the points P and Q such that $\triangle\mathrm{OPQ}$ is an isosceles triangle and $\angle\mathrm{POQ} = 90^\circ$. If $l = \mathrm{OP}^2 + \mathrm{PQ}^2 + \mathrm{QO}^2$, then the greatest integer less than or equal to l is :

(A) 42 (B) 46 (C) 48 (D) 44

Open solution

Q38. JEE Main 2024 (Online) 4th April Morning Shift

The vertices of a triangle are $\mathrm{A}(-1, 3), \mathrm{B}(-2, 2)$ and $\mathrm{C}(3, -1)$. A new triangle is formed by shifting the sides of the triangle by one unit inwards. Then the equation of the side of the new triangle nearest to origin is :

(A) $-x + y - (2 - \sqrt{2}) = 0$ (B) $x + y - (2 - \sqrt{2}) = 0$

(C) $x + y + (2 - \sqrt{2}) = 0$ (D) $x - y - (2 + \sqrt{2}) = 0$

Open solution

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Q39. JEE Main 2024 (Online) 31st January Evening Shift

Let $A(a,b), B(3,4)$ and $C(-6,-8)$ respectively denote the centroid, circumcentre and orthocentre of a triangle. Then, the distance of the point $P(2a+3, 7b+5)$ from the line $2x+3y-4=0$ measured parallel to the line $x-2y-1=0$ is

(A) $\frac{17\sqrt{5}}{6}$ (B) $\frac{15\sqrt{5}}{7}$ (C) $\frac{17\sqrt{5}}{7}$ (D) $\frac{\sqrt{5}}{17}$

Open solution

Q40. JEE Main 2024 (Online) 31st January Morning Shift

Let $\alpha, \beta, \gamma, \delta \in \mathbb{Z}$ and let $A(\alpha,\beta), B(1,0), C(\gamma,\delta)$ and $D(1,2)$ be the vertices of a parallelogram ABCD. If $AB=\sqrt{10}$ and the points A and C lie on the line $3y=2x+1$, then $2(\alpha+\beta+\gamma+\delta)$ is equal to

(A) 8 (B) 5 (C) 12 (D) 10

Open solution

Q41. JEE Main 2024 (Online) 30th January Evening Shift

If $x^2-y^2+2hxy+2gx+2fy+c=0$ is the locus of a point, which moves such that it is always equidistant from the lines $x+2y+7=0$ and $2x-y+8=0$, then the value of $g+c+h-f$ equals

(A) 8 (B) 14 (C) 29 (D) 6

Open solution

Q42. JEE Main 2024 (Online) 30th January Morning Shift

A line passing through the point $\mathrm{A}(9,0)$ makes an angle of 30° with the positive direction of x-axis. If this line is rotated about A through an angle of 15° in the clockwise direction, then its equation in the new position is :

(A) $\frac{y}{\sqrt{3}+2}+x=9$ (B) $\frac{x}{\sqrt{3}+2}+y=9$ (C) $\frac{x}{\sqrt{3}-2}+y=9$ (D) $\frac{y}{\sqrt{3}-2}+x=9$

Open solution

Q43. JEE Main 2024 (Online) 29th January Evening Shift

Let A be the point of intersection of the lines $3x+2y=14, 5x-y=6$ and B be the point of intersection of the lines $4x+3y=8, 6x+y=5$. The distance of the point $P(5,-2)$ from the line AB is

(A) $\frac{13}{2}$ (B) 8 (C) $\frac{5}{2}$ (D) 6

Open solution

Q44. JEE Main 2024 (Online) 29th January Evening Shift

The distance of the point $(2,3)$ from the line $2x-3y+28=0$, measured parallel to the line $\sqrt{3}x-y+1=0$, is equal to

(A) $3+4\sqrt{2}$ (B) $6\sqrt{3}$ (C) $4+6\sqrt{3}$ (D) $4\sqrt{2}$

Open solution

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Q45. JEE Main 2024 (Online) 29th January Morning Shift

In a $\triangle ABC$, suppose $y = x$ is the equation of the bisector of the angle B and the equation of the side AC is $2x - y = 2$. If $2AB = BC$ and the points A and B are respectively $(4, 6)$ and (α, β), then $\alpha + 2\beta$ is equal to

(A) 42 (B) 39 (C) 48 (D) 45

Open solution

Q46. JEE Main 2024 (Online) 27th January Evening Shift

Let R be the interior region between the lines $3x - y + 1 = 0$ and $x + 2y - 5 = 0$ containing the origin. The set of all values of a, for which the points $\left(a^2, a + 1\right)$ lie in R, is :

(A) $(-3, 0) \cup \left(\frac{2}{3}, 1\right)$ (B) $(-3, 0) \cup \left(\frac{1}{3}, 1\right)$

(C) $(-3, -1) \cup \left(\frac{1}{3}, 1\right)$ (D) $(-3, -1) \cup \left(-\frac{1}{3}, 1\right)$

Open solution

Q47. JEE Main 2024 (Online) 27th January Morning Shift

The portion of the line $4x + 5y = 20$ in the first quadrant is trisected by the lines L_1 and L_2 passing through the origin. The tangent of an angle between the lines L_1 and L_2 is :

(A) $\frac{30}{41}$ (B) $\frac{8}{5}$ (C) $\frac{2}{5}$ (D) $\frac{25}{41}$

Open solution

Q48. JEE Main 2023 (Online) 15th April Morning Shift

If (α, β) is the orthocenter of the triangle ABC with vertices $A(3, -7), B(-1, 2)$ and $C(4, 5)$, then $9\alpha - 6\beta + 60$ is equal to :

(A) 30 (B) 40 (C) 25 (D) 35

Open solution

Q49. JEE Main 2023 (Online) 13th April Evening Shift

Let (α, β) be the centroid of the triangle formed by the lines $15x - y = 82, 6x - 5y = -4$ and $9x + 4y = 17$. Then $\alpha + 2\beta$ and $2\alpha - \beta$ are the roots of the equation :

(A) $x^2 - 7x + 12 = 0$ (B) $x^2 - 13x + 42 = 0$ (C) $x^2 - 14x + 48 = 0$ (D) $x^2 - 10x + 25 = 0$

Open solution

Q50. JEE Main 2023 (Online) 12th April Morning Shift

If the point $\left(\alpha, \frac{7\sqrt{3}}{3}\right)$ lies on the curve traced by the mid-points of the line segments of the lines $x\cos\theta + y\sin\theta = 7, \theta \in \left(0, \frac{\pi}{2}\right)$ between the co-ordinates axes, then α is equal to :

(A) -7 (B) 7 (C) $-7\sqrt{3}$ (D) $7\sqrt{3}$

Open solution

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Q51. JEE Main 2023 (Online) 8th April Morning Shift

Open solution

Let $C(\alpha, \beta)$ be the circumcenter of the triangle formed by the lines

$4x + 3y = 69$

$4y - 3x = 17$, and

$x + 7y = 61$.

Then $(\alpha - \beta)^2 + \alpha + \beta$ is equal to :

(A) 15 (B) 17 (C) 16 (D) 18

Q52. JEE Main 2023 (Online) 6th April Morning Shift

Open solution

The straight lines l_1 and l_2 pass through the origin and trisect the line segment of the line L : $9x + 5y = 45$ between the axes. If m_1 and m_2 are the slopes of the lines l_1 and l_2, then the point of intersection of the line $y = (m_1 + m_2)x$ with L lies on :

(A) $6x - y = 15$ (B) $6x + y = 10$ (C) $y - x = 5$ (D) $y - 2x = 5$

Q53. JEE Main 2023 (Online) 1st February Morning Shift

Open solution

If the orthocentre of the triangle, whose vertices are (1, 2), (2, 3) and (3, 1) is (α, β), then the quadratic equation whose roots are $\alpha + 4\beta$ and $4\alpha + \beta$, is :

(A) $x^2 - 20x + 99 = 0$ (B) $x^2 - 22x + 120 = 0$

(C) $x^2 - 19x + 90 = 0$ (D) $x^2 - 18x + 80 = 0$

Q54. JEE Main 2023 (Online) 29th January Morning Shift

Open solution

Let B and C be the two points on the line $y + x = 0$ such that B and C are symmetric with respect to the origin. Suppose A is a point on $y - 2x = 2$ such that $\triangle ABC$ is an equilateral triangle. Then, the area of the $\triangle ABC$ is :

(A) $\frac{10}{\sqrt{3}}$ (B) $2\sqrt{3}$ (C) $3\sqrt{3}$ (D) $\frac{8}{\sqrt{3}}$

Q55. JEE Main 2023 (Online) 29th January Morning Shift

Open solution

A light ray emits from the origin making an angle 30° with the positive x-axis. After getting reflected by the line $x + y = 1$, if this ray intersects x-axis at Q, then the abscissa of Q is :

(A) $\frac{2}{\left(\sqrt{3}-1\right)}$ (B) $\frac{2}{3-\sqrt{3}}$

(C) $\frac{\sqrt{3}}{2\left(\sqrt{3}+1\right)}$ (D) $\frac{2}{3+\sqrt{3}}$

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Q56. JEE Main 2022 (Online) 29th July Evening Shift

Let m_1, m_2 be the slopes of two adjacent sides of a square of side a such that $a^2 + 11a + 3\left(m_1^2 + m_2^2\right) = 220$. If one vertex of the square is $(10(\cos\alpha - \sin\alpha), 10(\sin\alpha + \cos\alpha))$, where $\alpha \in \left(0, \frac{\pi}{2}\right)$ and the equation of one diagonal is $(\cos\alpha - \sin\alpha)x + (\sin\alpha + \cos\alpha)y = 10$, then $72\left(\sin^4\alpha + \cos^4\alpha\right) + a^2 - 3a + 13$ is equal to :

(A) 119 (B) 128 (C) 145 (D) 155

Open solution

Q57. JEE Main 2022 (Online) 29th July Evening Shift

Let $\mathrm{A}(\alpha, -2), \mathrm{B}(\alpha, 6)$ and $\mathrm{C}\left(\frac{\alpha}{4}, -2\right)$ be vertices of a $\triangle\mathrm{ABC}$. If $\left(5, \frac{\alpha}{4}\right)$ is the circumcentre of $\triangle\mathrm{ABC}$, then which of the following is NOT correct about $\triangle\mathrm{ABC}$?

(A) area is 24 (B) perimeter is 25

(C) circumradius is 5 (D) inradius is 2

Open solution

Q58. JEE Main 2022 (Online) 29th July Morning Shift

Let the circumcentre of a triangle with vertices A(a, 3), B(b, 5) and C(a, b), ab > 0 be P(1,1). If the line AP intersects the line BC at the point $\mathrm{Q}(k_1, k_2)$, then $k_1 + k_2$ is equal to :

(A) 2 (B) $\frac{4}{7}$ (C) $\frac{2}{7}$ (D) 4

Open solution

Q59. JEE Main 2022 (Online) 27th July Evening Shift

The equations of the sides AB, BC and CA of a triangle ABC are $2x + y = 0, x + \mathrm{p}y = 39$ and $x - y = 3$ respectively and $\mathrm{P}(2, 3)$ is its circumcentre. Then which of the following is NOT true?

(A) $(\mathrm{AC})^2 = 9\mathrm{p}$ (B) $(\mathrm{AC})^2 + \mathrm{p}^2 = 136$

(C) $32 < \text{area}\,(\Delta\mathrm{ABC}) < 36$ (D) $34 < \text{area}\,(\triangle\mathrm{ABC}) < 38$

Open solution

Q60. JEE Main 2022 (Online) 27th July Morning Shift

Let $A(1, 1), B(-4, 3), C(-2, -5)$ be vertices of a triangle ABC, P be a point on side BC, and Δ_1 and Δ_2 be the areas of triangles APB and ABC, respectively. If $\Delta_1 : \Delta_2 = 4 : 7$, then the area enclosed by the lines AP, AC and the x-axis is :

(A) $\frac{1}{4}$ (B) $\frac{3}{4}$ (C) $\frac{1}{2}$ (D) 1

Open solution

Q61. JEE Main 2022 (Online) 26th July Morning Shift

A point P moves so that the sum of squares of its distances from the points $(1, 2)$ and $(-2, 1)$ is 14. Let $f(x, y) = 0$ be the locus of P, which intersects the x-axis at the points A, B and the y-axis at the points C, D. Then the area of the quadrilateral ACBD is equal to :

(A) $\frac{9}{2}$ (B) $\frac{3\sqrt{17}}{2}$ (C) $\frac{3\sqrt{17}}{4}$ (D) 9

Open solution

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Q62. JEE Main 2022 (Online) 25th July Evening Shift

Let the point $P(\alpha, \beta)$ be at a unit distance from each of the two lines $L_1 : 3x - 4y + 12 = 0$, and $L_2 : 8x + 6y + 11 = 0$. If P lies below L_1 and above L_2, then $100(\alpha + \beta)$ is equal to :

(A) -14 (B) 42 (C) -22 (D) 14

Open solution

Q63. JEE Main 2022 (Online) 25th July Morning Shift

A line, with the slope greater than one, passes through the point $A(4, 3)$ and intersects the line $x - y - 2 = 0$ at the point B. If the length of the line segment AB is $\frac{\sqrt{29}}{3}$, then B also lies on the line :

(A) $2x + y = 9$ (B) $3x - 2y = 7$ (C) $x + 2y = 6$ (D) $2x - 3y = 3$

Open solution

Q64. JEE Main 2022 (Online) 30th June Morning Shift

Let α_1, α_2 ($\alpha_1 < \alpha_2$) be the values of α for the points $(\alpha, -3)$, (2, 0) and (1, α) to be collinear. Then the equation of the line, passing through (α_1, α_2) and making an angle of $\frac{\pi}{3}$ with the positive direction of the x-axis, is :

(A) $x - \sqrt{3}y - 3\sqrt{3} + 1 = 0$ (B) $\sqrt{3}x - y + \sqrt{3} + 3 = 0$

(C) $x - \sqrt{3}y + 3\sqrt{3} + 1 = 0$ (D) $\sqrt{3}x - y + \sqrt{3} - 3 = 0$

Open solution

Q65. JEE Main 2022 (Online) 29th June Evening Shift

The distance of the origin from the centroid of the triangle whose two sides have the equations $x - 2y + 1 = 0$ and $2x - y - 1 = 0$ and whose orthocenter is $\left(\frac{7}{3}, \frac{7}{3}\right)$ is :

(A) $\sqrt{2}$ (B) 2 (C) $2\sqrt{2}$ (D) 4

Open solution

Q66. JEE Main 2022 (Online) 29th June Morning Shift

The distance between the two points A and A' which lie on y = 2 such that both the line segments AB and A' B (where B is the point (2, 3)) subtend angle $\frac{\pi}{4}$ at the origin, is equal to :

(A) 10 (B) $\frac{48}{5}$ (C) $\frac{52}{5}$ (D) 3

Open solution

Q67. JEE Main 2022 (Online) 28th June Evening Shift

Let a triangle be bounded by the lines L_1 : 2x + 5y = 10; L_2 : $-$4x + 3y = 12 and the line L_3, which passes through the point P(2, 3), intersects L_2 at A and L_1 at B. If the point P divides the line-segment AB, internally in the ratio 1 : 3, then the area of the triangle is equal to :

(A) $\frac{110}{13}$ (B) $\frac{132}{13}$ (C) $\frac{142}{13}$ (D) $\frac{151}{13}$

Open solution

Q68. JEE Main 2022 (Online) 27th June Morning Shift

In an isosceles triangle ABC, the vertex A is (6, 1) and the equation of the base BC is 2x + y = 4. Let the point B lie on the line x + 3y = 7. If (α, β) is the centroid of ΔABC, then 15($\alpha + \beta$) is equal to :

(A) 39 (B) 41 (C) 51 (D) 63

Open solution

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Q69. JEE Main 2022 (Online) 26th June Morning Shift

Let R be the point (3, 7) and let P and Q be two points on the line x + y = 5 such that PQR is an equilateral triangle. Then the area of ΔPQR is :

(A) $\frac{25}{4\sqrt{3}}$ (B) $\frac{25\sqrt{3}}{2}$ (C) $\frac{25}{\sqrt{3}}$ (D) $\frac{25}{2\sqrt{3}}$

Open solution

Q70. JEE Main 2022 (Online) 24th June Evening Shift

Let the area of the triangle with vertices A(1, α), B(α, 0) and C(0, α) be 4 sq. units. If the points $(\alpha, -\alpha)$, $(-\alpha, \alpha)$ and (α^2, β) are collinear, then β is equal to :

(A) 64 (B) -8 (C) -64 (D) 512

Open solution

Q71. JEE Main 2021 (Online) 31st August Evening Shift

Let A be the set of all points (α, β) such that the area of triangle formed by the points (5, 6), (3, 2) and (α, β) is 12 square units. Then the least possible length of a line segment joining the origin to a point in A, is :

(A) $\frac{4}{\sqrt{5}}$ (B) $\frac{16}{\sqrt{5}}$ (C) $\frac{8}{\sqrt{5}}$ (D) $\frac{12}{\sqrt{5}}$

Open solution

Q72. JEE Main 2021 (Online) 31st August Morning Shift

If p and q are the lengths of the perpendiculars from the origin on the lines,

x cosec α $-$ y sec α = k cot 2α and

x sinα + y cosα = k sin2α

respectively, then k^2 is equal to :

(A) $4p^2 + q^2$ (B) $2p^2 + q^2$ (C) $p^2 + 2q^2$ (D) $p^2 + 4q^2$

Open solution

Q73. JEE Main 2021 (Online) 27th August Morning Shift

Let A be a fixed point (0, 6) and B be a moving point (2t, 0). Let M be the mid-point of AB and the perpendicular bisector of AB meets the y-axis at C. The locus of the mid-point P of MC is :

(A) $3x^2 - 2y - 6 = 0$ (B) $3x^2 + 2y - 6 = 0$ (C) $2x^2 + 3y - 9 = 0$ (D) $2x^2 - 3y + 9 = 0$

Open solution

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Q74. JEE Main 2021 (Online) 27th July Evening Shift

The point P (a, b) undergoes the following three transformations successively :

(a) reflection about the line y = x.

(b) translation through 2 units along the positive direction of x-axis.

(c) rotation through angle $\frac{\pi}{4}$ about the origin in the anti-clockwise direction.

If the co-ordinates of the final position of the point P are $\left(-\frac{1}{\sqrt{2}}, \frac{7}{\sqrt{2}}\right)$, then the value of 2a + b is equal to :

(A) 13 (B) 9 (C) 5 (D) 7

Open solution

Q75. JEE Main 2021 (Online) 27th July Evening Shift

Two sides of a parallelogram are along the lines 4x + 5y = 0 and 7x + 2y = 0. If the equation of one of the diagonals of the parallelogram is 11x + 7y = 9, then other diagonal passes through the point :

(A) (1, 2) (B) (2, 2) (C) (2, 1) (D) (1, 3)

Open solution

Q76. JEE Main 2021 (Online) 18th March Evening Shift

Let the centroid of an equilateral triangle ABC be at the origin. Let one of the sides of the equilateral triangle be along the straight line x + y = 3. If R and r be the radius of circumcircle and incircle respectively of ΔABC, then (R + r) is equal to :

(A) $7\sqrt{2}$ (B) $\frac{9}{\sqrt{2}}$ (C) $2\sqrt{2}$ (D) $3\sqrt{2}$

Open solution

Q77. JEE Main 2021 (Online) 18th March Morning Shift

The number of integral values of m so that the abscissa of point of intersection of lines 3x + 4y = 9 and y = mx + 1 is also an integer, is :

(A) 1 (B) 2 (C) 3 (D) 0

Open solution

Q78. JEE Main 2021 (Online) 18th March Morning Shift

The equation of one of the straight lines which passes through the point (1, 3) and makes an angles $\tan^{-1}\left(\sqrt{2}\right)$ with the straight line, y + 1 = $3\sqrt{2}$ x is :

(A) $4\sqrt{2}x + 5y - \left(15 + 4\sqrt{2}\right) = 0$ (B) $5\sqrt{2}x + 4y - \left(15 + 4\sqrt{2}\right) = 0$

(C) $4\sqrt{2}x + 5y - 4\sqrt{2} = 0$ (D) $4\sqrt{2}x - 5y - \left(5 + 4\sqrt{2}\right) = 0$

Open solution

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Q79. JEE Main 2021 (Online) 17th March Morning Shift

In a triangle PQR, the co-ordinates of the points P and Q are $(-2, 4)$ and $(4, -2)$ respectively. If the equation of the perpendicular bisector of PR is $2x - y + 2 = 0$, then the centre of the circumcircle of the ΔPQR is :

Open solution

(A) $(-1, 0)$ (B) $(1, 4)$ (C) $(0, 2)$ (D) $(-2, -2)$

Q80. JEE Main 2021 (Online) 16th March Evening Shift

Let A$(-1, 1)$, B$(3, 4)$ and C$(2, 0)$ be given three points.
A line $y = mx$, $m > 0$, intersects lines AC and BC at point P and Q respectively. Let A_1 and A_2 be the areas of ΔABC and ΔPQC respectively, such that $A_1 = 3A_2$, then the value of m is equal to :

Open solution

(A) 1 (B) 3 (C) 2 (D) $\frac{4}{15}$

Q81. JEE Main 2021 (Online) 26th February Morning Shift

The intersection of three lines $x - y = 0$, $x + 2y = 3$ and $2x + y = 6$ is a :

Open solution

(A) Right angled triangle (B) Equilateral triangle
(C) None of the above (D) Isosceles triangle

Q82. JEE Main 2021 (Online) 25th February Morning Shift

The image of the point (3, 5) in the line $x - y + 1 = 0$, lies on :

Open solution

(A) $(x - 4)^2 + (y - 4)^2 = 8$ (B) $(x - 4)^2 + (y + 2)^2 = 16$
(C) $(x - 2)^2 + (y - 2)^2 = 12$ (D) $(x - 2)^2 + (y - 4)^2 = 4$

Q83. JEE Main 2021 (Online) 24th February Morning Shift

A man is walking on a straight line. The arithmetic mean of the reciprocals of the intercepts of this line on the coordinate axes is $\frac{1}{4}$. Three stones A, B and C are placed at the points (1, 1), (2, 2) and (4, 4) respectively. Then, which of these stones is / are on the path of the man?

Open solution

(A) A only (B) All the three (C) C only (D) B only

Q84. JEE Main 2020 (Online) 6th September Evening Slot

Let L denote the line in the xy-plane with x and y intercepts as 3 and 1 respectively. Then the image of the point $(-1, -4)$ in this line is :

Open solution

(A) $\left(\frac{11}{5}, \frac{28}{5}\right)$ (B) $\left(\frac{29}{5}, \frac{11}{5}\right)$
(C) $\left(\frac{29}{5}, \frac{8}{5}\right)$ (D) $\left(\frac{8}{5}, \frac{29}{5}\right)$

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Q85. JEE Main 2020 (Online) 6th September Morning Slot

A ray of light coming from the point $(2, 2\sqrt{3})$ is incident at an angle 30° on the line x = 1 at the point A. The ray gets reflected on the line x = 1 and meets x-axis at the point B. Then, the line AB passes through the point :

Open solution

(A) $(3, -\sqrt{3})$ (B) $(4, -\sqrt{3})$

(C) $\left(4, -\frac{\sqrt{3}}{2}\right)$ (D) $\left(3, -\frac{1}{\sqrt{3}}\right)$

Q86. JEE Main 2020 (Online) 4th September Evening Slot

If the perpendicular bisector of the line segment joining the points P(1 ,4) and Q(k, 3) has y-intercept equal to −4, then a value of k is :

Open solution

(A) $\sqrt{14}$ (B) -4 (C) −2 (D) $\sqrt{15}$

Q87. JEE Main 2020 (Online) 3rd September Evening Slot

If a ΔABC has vertices A(−1, 7), B(−7, 1) and C(5, −5), then its orthocentre has coordinates :

Open solution

(A) (−3, 3) (B) (3, −3)

(C) $\left(\frac{3}{5}, -\frac{3}{5}\right)$ (D) $\left(-\frac{3}{5}, \frac{3}{5}\right)$

Q88. JEE Main 2020 (Online) 2nd September Evening Slot

The set of all possible values of θ in the interval $(0, \pi)$ for which the points (1, 2) and (sin θ, cos θ) lie on the same side of the line x + y = 1 is :

Open solution

(A) $\left(0, \frac{\pi}{4}\right)$ (B) $\left(0, \frac{3\pi}{4}\right)$

(C) $\left(\frac{\pi}{4}, \frac{3\pi}{4}\right)$ (D) $\left(0, \frac{\pi}{2}\right)$

Q89. JEE Main 2020 (Online) 9th January Morning Slot

Let C be the centroid of the triangle with vertices (3, −1), (1, 3) and (2, 4). Let P be the point of intersection of the lines x + 3y − 1 = 0 and 3x − y + 1 = 0. Then the line passing through the points C and P also passes through the point :

Open solution

(A) (−9, −7) (B) (9, 7) (C) (7, 6) (D) (−9, −6)

Q90. JEE Main 2020 (Online) 8th January Morning Slot

Let two points be A(1, −1) and B(0, 2). If a point P(x', y') be such that the area of ΔPAB = 5 sq. units and it lies on the line, 3x + y − 4λ = 0, then a value of λ is :

Open solution

(A) 4 (B) 1 (C) -3 (D) 3

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Q91. JEE Main 2020 (Online) 7th January Evening Slot

The locus of the mid-points of the perpendiculars drawn from points on the line, x = 2y to the line x = y is :

(A) 3x - 2y = 0 (B) 7x - 5y = 0 (C) 2x - 3y = 0 (D) 5x - 7y = 0

Open solution

Q92. JEE Main 2019 (Online) 12th April Evening Slot

A straight line L at a distance of 4 units from the origin makes positive intercepts on the coordinate axes and the perpendicular from the origin to this line makes an angle of 60° with the line x + y = 0. Then an equation of the line L is :

(A) $x + \sqrt{3}y = 8$ (B) $\sqrt{3}x + y = 8$

(C) $(\sqrt{3} + 1)x + (\sqrt{3} - 1)y = 8\sqrt{2}$ (D) $(\sqrt{3} - 1)x + (\sqrt{3} + 1)y = 8\sqrt{2}$

Open solution

Q93. JEE Main 2019 (Online) 10th April Evening Slot

Lines are drawn parallel to the line 4x – 3y + 2 = 0, at a distance $\frac{3}{5}$ from the origin. Then which one of the following points lies on any of these lines ?

(A) $\left(\frac{1}{4}, -\frac{1}{3}\right)$ (B) $\left(-\frac{1}{4}, \frac{2}{3}\right)$

(C) $\left(-\frac{1}{4}, -\frac{2}{3}\right)$ (D) $\left(\frac{1}{4}, \frac{1}{3}\right)$

Open solution

Q94. JEE Main 2019 (Online) 10th April Morning Slot

The region represented by$|x - y| \leq 2$ and $|x + y| \leq 2$ is bounded by a :

(A) rhombus of area $8\sqrt{2}$ sq. units (B) square of side length $2\sqrt{2}$ units

(C) square of area 16 sq. units (D) rhombus of side length 2 units

Open solution

Q95. JEE Main 2019 (Online) 9th April Evening Slot

If the two lines x + (a – 1) y = 1 and $2x + a^2y = 1$ ($a \in R - \{0, 1\}$) are perpendicular, then the distance of their point of intersection from the origin is :

(A) $\frac{2}{\sqrt{5}}$ (B) $\frac{\sqrt{2}}{5}$ (C) $\frac{2}{5}$ (D) $\sqrt{\frac{2}{5}}$

Open solution

Q96. JEE Main 2019 (Online) 9th April Morning Slot

Slope of a line passing through P(2, 3) and intersecting the line, x + y = 7 at a distance of 4 units from P, is :

(A) $\frac{\sqrt{7}-1}{\sqrt{7}+1}$ (B) $\frac{\sqrt{5}-1}{\sqrt{5}+1}$

(C) $\frac{1-\sqrt{5}}{1+\sqrt{5}}$ (D) $\frac{1-\sqrt{7}}{1+\sqrt{7}}$

Open solution

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Q97. JEE Main 2019 (Online) 8th April Evening Slot

Suppose that the points (h,k), (1,2) and (−3,4) lie on the line L_1 . If a line L_2 passing through the points (h,k) and (4,3) is perpendicular to L_1 , then $\frac{k}{h}$ equals :

(A) $\frac{1}{3}$ (B) 3 (C) 0 (D) $-\frac{1}{7}$

Open solution

Q98. JEE Main 2019 (Online) 8th April Morning Slot

A point on the straight line, 3x + 5y = 15 which is equidistant from the coordinate axes will lie only in :

(A) 1st and 2nd qudratants (B) 4th qudratant

(C) 1st and 2nd and 4th qudratants (D) 1st qudratant

Open solution

Q99. JEE Main 2019 (Online) 8th April Morning Slot

Let O(0, 0) and A(0, 1) be two fixed points. Then the locus of a point P such that the perimeter of ΔAOP is 4, is :

(A) $9x^2 + 8y^2 - 8y = 16$ (B) $8x^2 - 9y^2 + 9y = 18$

(C) $8x^2 + 9y^2 - 9y = 18$ (D) $9x^2 - 8y^2 + 8y = 16$

Open solution

Q100. JEE Main 2019 (Online) 12th January Evening Slot

If a straight line passing through the point P(−3, 4) is such that its intercepted portion between the coordinate axes is bisected at P, then its equation is :

(A) x − y + 7 = 0 (B) 4x − 3y + 24 = 0 (C) 4x + 3y = 0 (D) 3x − 4y + 25 = 0

Open solution

Q101. JEE Main 2019 (Online) 12th January Morning Slot

If the straight line, 2x − 3y + 17 = 0 is perpendicular to the line passing through the points (7, 17) and (15, β), then β equals :

(A) $\frac{35}{3}$ (B) -5 (C) $-\frac{35}{3}$ (D) 5

Open solution

Q102. JEE Main 2019 (Online) 11th January Evening Slot

If in a parallelogram ABDC, the coordinates of A, B and C are respectively (1, 2), (3, 4) and (2, 5), then the equation of the diagonal AD is :

(A) 5x + 3y − 11 = 0 (B) 5x − 3y + 1 = 0 (C) 3x − 5y + 7 = 0 (D) 3x + 5y − 13 = 0

Open solution

Q103. JEE Main 2019 (Online) 10th January Evening Slot

Two sides of a parallelogram are along the lines, x + y = 3 & x − y + 3 = 0. If its diagonals intersect at (2, 4), then one of its vertex is :

(A) (2, 1) (B) (2, 6) (C) (3, 5) (D) (3, 6)

Open solution

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Q104. JEE Main 2019 (Online) 10th January Evening Slot

Two vertices of a triangle are (0, 2) and (4, 3). If its orthocenter is at the origin, then its third vertex lies in which quadrant :

(A) third (B) fourth (C) second (D) first

Open solution

Q105. JEE Main 2019 (Online) 10th January Morning Slot

A point P moves on the line $2x - 3y + 4 = 0$. If Q(1, 4) and R (3, – 2) are fixed points, then the locus of the centroid of ΔPQR is a line :

(A) parallel to y-axis (B) with slope $\frac{2}{3}$

(C) parallel to x-axis (D) with slope $\frac{3}{2}$

Open solution

Q106. JEE Main 2019 (Online) 10th January Morning Slot

If the line $3x + 4y - 24 = 0$ intersects the x-axis at the point A and the y-axis at the point B, then the incentre of the triangle OAB, where O is the origin, is :

(A) (3, 4) (B) (2, 2) (C) (4, 4) (D) (4, 3)

Open solution

Q107. JEE Main 2019 (Online) 9th January Evening Slot

Let the equations of two sides of a triangle be $3x - 2y + 6 = 0$ and $4x + 5y - 20 = 0$. If the orthocentre of this triangle is at (1, 1), then the equation of its third side is :

(A) $122y - 26x - 1675 = 0$ (B) $122y + 26x + 1675 = 0$

(C) $26x + 61y + 1675 = 0$ (D) $26x - 122y - 1675 = 0$

Open solution

Q108. JEE Main 2018 (Offline)

A straight line through a fixed point (2, 3) intersects the coordinate axes at distinct points P and Q. If O is the origin and the rectangle OPRQ is completed, then the locus of R is :

(A) $3x + 2y = 6xy$ (B) $3x + 2y = 6$ (C) $2x + 3y = xy$ (D) $3x + 2y = xy$

Open solution

Q109. JEE Main 2018 (Online) 15th April Evening Slot

The sides of a rhombus ABCD are parallel to the lines, $x - y + 2 = 0$ and $7x - y + 3 = 0$. If the diagonals of the rhombus intersect at P(1, 2) and the vertex A (different from the origin) is on the y-axis, then the ordinate of A is :

(A) $\frac{5}{2}$ (B) $\frac{7}{4}$ (C) 2 (D) $\frac{7}{2}$

Open solution

Q110. JEE Main 2018 (Online) 15th April Evening Slot

The foot of the perpendicular drawn from the origin, on the line, $3x + y = \lambda$ ($\lambda \neq 0$) is P. If the line meets x-axis at A and y-axis at B, then the ratio BP : PA is :

(A) 1 : 3 (B) 3 : 1 (C) 1 : 9 (D) 9 : 1

Open solution

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Q111. JEE Main 2018 (Online) 15th April Morning Slot

In a triangle ABC, coordinates of A are (1, 2) and the equations of the medians through B and C are respectively, x + y = 5 and x = 4. Then area of Δ ABC (in sq. units) is :

(A) 12 (B) 4 (C) 5 (D) 9

Open solution

Q112. JEE Main 2017 (Online) 9th April Morning Slot

A square, of each side 2, lies above the x-axis and has one vertex at the origin. If one of the sides passing through the origin makes an angle 30° with the positive direction of the x-axis, then the sum of the x-coordinates of the vertices of the square is :

(A) $2\sqrt{3}-1$ (B) $2\sqrt{3}-2$ (C) $\sqrt{3}-2$ (D) $\sqrt{3}-1$

Open solution

Q113. JEE Main 2017 (Offline)

Let k be an integer such that the triangle with vertices (k, – 3k), (5, k) and (–k, 2) has area 28 sq. units. Then the orthocentre of this triangle is at the point :

(A) $\left(1, \frac{3}{4}\right)$ (B) $\left(1, -\frac{3}{4}\right)$
(C) $\left(2, \frac{1}{2}\right)$ (D) $\left(2, -\frac{1}{2}\right)$

Open solution

Q114. JEE Main 2016 (Online) 10th April Morning Slot

A ray of light is incident along a line which meets another line, 7x – y + 1 = 0, at the point (0, 1). The ray is then reflected from this point along the line, y + 2x = 1. Then the equation of the line of incidence of the ray of light is :

(A) 41x – 38y + 38 = 0 (B) 41x + 25y – 25 = 0
(C) 41x + 38y – 38 = 0 (D) 41x – 25y + 25 = 0

Open solution

Q115. JEE Main 2016 (Online) 10th April Morning Slot

A straight line through origin O meets the lines 3y = 10 – 4x and 8x + 6y + 5 = 0 at points A and B respectively. Then O divides the segment AB in the ratio :

(A) 2 : 3 (B) 1 : 2 (C) 4 : 1 (D) 3 : 4

Open solution

Q116. JEE Main 2016 (Online) 9th April Morning Slot

If a variable line drawn through the intersection of the lines $\frac{x}{3}+\frac{y}{4}=1$ and $\frac{x}{4}+\frac{y}{3}=1$, meets the coordinate axes at A and B, (A $\neq$ B), then the locus of the midpoint of AB is :

(A) 6xy = 7(x + y) (B) $4(x+y)^2 - 28(x+y) + 49 = 0$
(C) 7xy = 6(x + y) (D) $14(x+y)^2 - 97(x+y) + 168 = 0$

Open solution

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Q117. JEE Main 2016 (Online) 9th April Morning Slot

The point (2, 1) is translated parallel to the line L : x− y = 4 by $2\sqrt{3}$ units. If the newpoint Q lies in the third quadrant, then the equation of the line passing through Q and perpendicular to L is :

(A) x + y = 2 − $\sqrt{6}$

(B) x + y = 3 − $3\sqrt{6}$

(C) x + y = 3 − $2\sqrt{6}$

(D) 2x + 2y = 1 − $\sqrt{6}$

Open solution

Q118. JEE Main 2016 (Offline)

Two sides of a rhombus are along the lines, $x - y + 1 = 0$ and $7x - y - 5 = 0$. If its diagonals intersect at $(-1, -2)$, then which one of the following is a vertex of this rhombus?

(A) $\left(\frac{1}{3}, -\frac{8}{3}\right)$

(B) $\left(-\frac{10}{3}, -\frac{7}{3}\right)$

(C) $(-3, -9)$

(D) $(-3, -8)$

Open solution

Q119. JEE Main 2015 (Online) 11th April Morning Slot

A straight line L through the point (3, − 2) is inclined at an angle of 60° to the line $\sqrt{3}x + y = 1$. If L also intersects the x-axis, then the equation of L is :

(A) y + $\sqrt{3}$ x + 2 − $3\sqrt{3}$ = 0

(B) y − $\sqrt{3}$ x + 2 + $3\sqrt{3}$ = 0

(C) $\sqrt{3}$ y − x + 3 + $2\sqrt{3}$ = 0

(D) $\sqrt{3}$ y + x − 3 + $2\sqrt{3}$ = 0

Open solution

Q120. JEE Main 2015 (Online) 10th April Morning Slot

The points $\left(0, \frac{8}{3}\right)$, (1, 3) and (82, 30) :

(A) form an obtuse angled triangle

(B) form an acute angled triangle

(C) form a right angled triangle.

(D) lie on a straight line.

Open solution

Q121. JEE Main 2015 (Online) 10th April Morning Slot

Let L be the line passing through the point P(1, 2) such that its intercepted segment between the co-ordinate axes is bisected at P. If L_1 is the line perpendicular to L and passing through the point (−2, 1), then the point of intersection of L and L_1 is :

(A) $\left(\frac{4}{5}, \frac{12}{5}\right)$

(B) $\left(\frac{11}{20}, \frac{29}{10}\right)$

(C) $\left(\frac{3}{10}, \frac{17}{5}\right)$

(D) $\left(\frac{3}{5}, \frac{23}{10}\right)$

Open solution

Q122. JEE Main 2015 (Offline)

The number of points, having both co-ordinates as integers, that lie in the interior of the triangle with vertices $(0, 0)$ $(0, 41)$ and $(41, 0)$ is :

(A) 820 (B) 780 (C) 901 (D) 861

Open solution

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Q123. JEE Main 2014 (Online) 19th April Morning Slot

The circumcenter of a triangle lies at the origin and its centroid is the mid point of the line segment joining the points $(a^2 + 1, a^2 + 1)$ and $(2a, -2a)$, $a \neq 0$. Then for any a, the orthocenter of this triangle lies on the line :

(A) $y - 2ax = 0$

(B) $y - (a^2 + 1)x = 0$

(C) $y + x = 0$

(D) $(a - 1)^2x - (a + 1)^2 y = 0$

Open solution

Q124. JEE Main 2014 (Online) 19th April Morning Slot

If a line L is perpendicular to the line $5x - y = 1$, and the area of the triangle formed by the line L and the coordinate axes is 5, then the distance of line L from the line $x + 5y = 0$ is :

(A) $\frac{7}{\sqrt{5}}$

(B) $\frac{5}{\sqrt{13}}$

(C) $\frac{7}{\sqrt{13}}$

(D) $\frac{5}{\sqrt{7}}$

Open solution

Q125. JEE Main 2014 (Online) 12th April Morning Slot

If a line intercepted between the coordinate axes is trisected at a point $A(4, 3)$, which is nearer to x-axis, then its equation is :

(A) $4x - 3y = 7$

(B) $3x + 2y = 18$

(C) $3x + 8y = 36$

(D) $x + 3y = 13$

Open solution

Q126. JEE Main 2014 (Online) 12th April Morning Slot

If the three distinct lines $x + 2ay + a = 0$, $x + 3by + b = 0$ and $x + 4ay + a = 0$ are concurrent, then the point (a, b) lies on a :

(A) circle

(B) hyperbola

(C) straight line

(D) parabola

Open solution

Q127. JEE Main 2014 (Online) 11th April Morning Slot

The base of an equilateral triangle is along the line given by $3x + 4y = 9$. If a vertex of the triangle is (1, 2), then length of a side of the triangle is :

(A) $\frac{2\sqrt{3}}{15}$

(B) $\frac{4\sqrt{3}}{15}$

(C) $\frac{4\sqrt{3}}{5}$

(D) $\frac{2\sqrt{3}}{5}$

Open solution

Q128. JEE Main 2014 (Online) 9th April Morning Slot

Let a and b be any two numbers satisfying $\frac{1}{a^2} + \frac{1}{b^2} = \frac{1}{4}$. Then, the foot of perpendicular from the origin on the variable line, $\frac{x}{a} + \frac{y}{b} = 1$, lies on :

(A) a hyperbola with each semi-axis = $\sqrt{2}$.

(B) a hyperbola with each semi-axis = 2.

(C) a circle of radius = 2

(D) a circle of radius = $\sqrt{2}$.

Open solution

Q129. JEE Main 2014 (Online) 9th April Morning Slot

Given three points P, Q, R with P(5, 3) and R lies on the x-axis. If equation of RQ is $x - 2y = 2$ and PQ is parallel to the x-axis, then the centroid of ΔPQR lies on the line :

(A) $2x + y - 9 = 0$

(B) $x - 2y + 1 = 0$

(C) $5x - 2y = 0$

(D) $2x - 5y = 0$

Open solution

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Q130. JEE Main 2014 (Offline)

Let PS be the median of the triangle with vertices $P(2,2)$, $Q(6,-1)$ and $R(7,3)$. The equation of the line passing through $(1,-1)$ and parallel to PS is :

(A) $4x+7y+3=0$ (B) $2x-9y-11=0$ (C) $4x-7y-11=0$ (D) $2x+9y+7=0$

Open solution

Q131. JEE Main 2014 (Offline)

Let a, b, c and d be non-zero numbers. If the point of intersection of the lines $4ax+2ay+c=0$ and $5bx+2by+d=0$ lies in the fourth quadrant and is equidistant from the two axes then :

(A) $3bc-2ad=0$ (B) $3bc+2ad=0$ (C) $2bc-3ad=0$ (D) $2bc+3ad=0$

Open solution

Q132. JEE Main 2013 (Online) 25th April Morning Slot

Let A(-3, 2) and B(-2, 1) be the vertices of a triangle ABC. If the centroid of this triangle lies on the line 3x + 4y + 2 = 0, then the vertex C lies on the line:

(A) 4x + 3y + 5 = 0 (B) 3x + 4y + 3 = 0 (C) 4x + 3y + 3 = 0 (D) 3x + 4y + 5 = 0

Open solution

Q133. JEE Main 2013 (Online) 25th April Morning Slot

If the image of point P(2, 3) in a line L is Q(4, 5), then the image of point R(0, 0) in the same line is:

(A) (2 , 2) (B) (4, 5) (C) (3, 4) (D) (7, 7)

Open solution

Q134. JEE Main 2013 (Online) 23th April Morning Slot

If the extremities of the base of an isosceles triangle are the points (2a, 0) and (0, a) and the equation of one of the sides is x = 2a, then the area of the triangle, in square units, is

(A) $\frac{5}{4}a^2$ (B) $\frac{5}{2}a^2$ (C) $\frac{25}{4}a^2$ (D) $5a^2$

Open solution

Q135. JEE Main 2013 (Online) 23th April Morning Slot

Let θ_1 be the angle between two lines 2x + 3y + c_1 = 0 and -x + 5y + c_2 = 0 and θ_2 be the angle between two lines 2x + 3y + c_1 = 0 and -x + 5y + c_3 = 0, where c_1, c_2, c_3 are any real numbers:

Open solution

Statement 1: If c_2 and c_3 are proportional, then $\theta_1 = \theta_2$

Statement 2: $\theta_1 = \theta_2$ for all c_2 and c_3

(A) Statement 1 is true; Statement 2 is true; Statement 2 is a correct explanation of Statement 1.

(B) Statement 1 is true; Statement 2 is true; Statement 2 is not a correct explanation of Statement 1.

(C) Statement 1 is false; Statement 2 is true;

(D) Statement 1 is true; Statement 2 is false;

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Q136. JEE Main 2013 (Online) 22th April Morning Slot

If the x-intercept of some line L is double as that of the line, 3x + 4y = 12 and the y-intercept of L is half as that of the same line, then the slope of L is:

(A) -3 (B) -3/8 (C) -3/2 (D) -3/16

Open solution

Q137. JEE Main 2013 (Online) 9th April Morning Slot

A light ray emerging from the point source placed at P(1, 3) is reflected at a point Q in the axis of x. If the reflected ray passes through the point R(6,7), then the abscissa of Q is:

(A) 1 (B) 3 (C) $\frac{7}{2}$ (D) $\frac{5}{2}$

Open solution

Q138. JEE Main 2013 (Online) 9th April Morning Slot

If the three lines x − 3y = p, ax + 2y = q and ax + y = r form a right-angled triangle then:

(A) $a^2 - 9a + 18 = 0$ (B) $a^2 - 6a - 12 = 0$ (C) $a^2 - 6a - 18 = 0$ (D) $a^2 - 9a + 12 = 0$

Open solution

Q139. JEE Main 2013 (Offline)

A ray of light along $x + \sqrt{3}y = \sqrt{3}$ gets reflected upon reaching X-axis, the equation of the reflected ray is :

(A) $y = x + \sqrt{3}$ (B) $\sqrt{3}y = x - \sqrt{3}$ (C) $y = \sqrt{3}x - \sqrt{3}$ (D) $\sqrt{3}y = x - 1$

Open solution

Q140. JEE Main 2013 (Offline)

The x-coordinate of the incentre of the triangle that has the coordinates of mid points of its sides as $(0, 1)(1, 1)$ and $(1, 0)$ is :

(A) $2 + \sqrt{2}$ (B) $2 - \sqrt{2}$ (C) $1 + \sqrt{2}$ (D) $1 - \sqrt{2}$

Open solution

Q141. AIEEE 2012 (Online) 26th May

The line parallel to x-axis and passing through the point of intersection of lines $ax + 2by + 3b = 0$ and $bx - 2ay - 3a = 0$, where $(a, b) \neq (0, 0)$ is

(A) above x-axis at a distance $2/3$ from it (B) above x-axis at a distance $3/2$ from it

(C) below x-axis at a distance $3/2$ from it (D) below x-axis at a distance $2/3$ from it

Open solution

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Q142. AIEEE 2012 (Online) 26th May

Consider the straight lines
$L_1 : x - y = 1$
$L_2 : x + y = 1$
$L_3 : 2x + 2y = 5$
$L_4 : 2x - 2y = 7$
The correct statement is

(A) $L_1 \parallel L_4$, $L_2 \parallel L_3$, L_1 intersect L_4.
(B) $L_1 \perp L_2$, $L_1 \parallel L_3$, L_1 intersect L_2.
(C) $L_1 \perp L_2$, $L_2 \parallel L_3$, L_1 intersect L_4.
(D) $L_1 \perp L_2$, $L_1 \perp L_3$, L_2 intersect L_4.

Open solution

Q143. AIEEE 2012 (Online) 19th May

Let L be the line $y = 2x$, in the two dimensional plane.
Statement 1: The image of the point $(0, 1)$ in L is the point $\left(\frac{4}{5}, \frac{3}{5}\right)$.
Statement 2: The points $(0, 1)$ and $\left(\frac{4}{5}, \frac{3}{5}\right)$ lie on opposite sides of the line L and are at equal distance from it.

(A) Statement 1 is true, Statement 2 is false.
(B) Statement 1 is true, Statement 2 is true, Statement 2 is not a correct explanation for Statement 1.
(C) Statement 1 is true, Statement 2 is true, Statement 2 is a correct explanation for Statement 1.
(D) Statement 1 is false, Statement 2 is true.

Open solution

Q144. AIEEE 2012 (Online) 12th May

If the point $(1, a)$ lies between the straight lines $x + y = 1$ and $2(x + y) = 3$ then a lies in interval

(A) $\left(\frac{3}{2}, \infty\right)$ (B) $\left(1, \frac{3}{2}\right)$ (C) $(-\infty, 0)$ (D) $\left(0, \frac{1}{2}\right)$

Open solution

Q145. AIEEE 2012 (Online) 12th May

If two vertices of a triangle are $(5, -1)$ and $(-2, 3)$ and its orthocentre is at $(0, 0)$, then the third vertex is

(A) $(4, -7)$ (B) $(-4, -7)$ (C) $(-4, 7)$ (D) $(4, 7)$

Open solution

Q146. AIEEE 2012 (Online) 7th May

If the straight lines $x + 3y = 4$, $3x + y = 4$ and $x + y = 0$ form a triangle, then the triangle is

(A) scalene
(B) equilateral triangle
(C) isosceles
(D) right angled isosceles

Open solution

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Q147. AIEEE 2012 (Online) 7th May

The point of intersection of the lines $(a^3+3)x+ay+a-3=0$ and $(a^5+2)x+(a+2)y+2a+3=0$ (a real) lies on the y-axis for

(A) no value of a
(B) more than two values of a
(C) exactly one value of a
(D) exactly two values of a

Open solution

Q148. AIEEE 2012

If the line $2x+y=k$ passes through the point which divides the line segment joining the points $(1,1)$ and $(2,4)$ in the ratio $3:2$, then k equals :

(A) $\frac{29}{5}$ (B) 5 (C) 6 (D) $\frac{11}{5}$

Open solution

Q149. AIEEE 2010

The line L given by $\frac{x}{5}+\frac{y}{b}=1$ passes through the point $(13,32)$. The line K is parrallel to L and has the equation $\frac{x}{c}+\frac{y}{3}=1$. Then the distance between L and K is :

(A) $\sqrt{17}$ (B) $\frac{17}{\sqrt{15}}$ (C) $\frac{23}{\sqrt{17}}$ (D) $\frac{23}{\sqrt{15}}$

Open solution

Q150. AIEEE 2009

The shortest distance between the line $y-x=1$ and the curve $x=y^2$ is :

(A) $\frac{2\sqrt{3}}{8}$ (B) $\frac{3\sqrt{2}}{5}$ (C) $\frac{\sqrt{3}}{4}$ (D) $\frac{3\sqrt{2}}{8}$

Open solution

Q151. AIEEE 2009

The lines $p(p^2+1)x-y+q=0$ and $(p^2+1)^2x+(p^2+1)y+2q=0$ are perpendicular to a common line for :

(A) exactly one values of p
(B) exactly two values of p
(C) more than two values of p
(D) no value of p

Open solution

Q152. AIEEE 2008

The perpendicular bisector of the line segment joining P(1, 4) and Q(k, 3) has y-intercept -4. Then a possible value of k is :

(A) 1 (B) 2 (C) -2 (D) -4

Open solution

Q153. AIEEE 2007

Let A (h,k), B$(1,1)$ and C $(2,1)$ be the vertices of a right angled triangle with AC as its hypotenuse. If the area of the triangle is 1 square unit, then the set of values which $'k'$ can take is given by :

(A) $\{-1,3\}$ (B) $\{-3,-2\}$ (C) $\{1,3\}$ (D) $\{0,2\}$

Open solution

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Q154. AIEEE 2006

A straight line through the point $A(3,4)$ is such that its intercept between the axes is bisected at A. Its equation is :

(A) $x+y=7$ (B) $3x-4y+7=0$ (C) $4x+3y=24$ (D) $3x+4y=25$

Open solution

Q155. AIEEE 2006

If (a,a^2) falls inside the angle made by the lines $y=\frac{x}{2}$, $x>0$ and $y=3x$, $x>0$, then a belong to :

(A) $(0,\frac{1}{2})$ (B) $(3,\infty)$ (C) $(\frac{1}{2},3)$ (D) $(-3,-\frac{1}{2})$

Open solution

Q156. AIEEE 2005

The line parallel to the x - axis and passing through the intersection of the lines $ax+2by+3b=0$ and $bx-2ay-3a=0$, where $(a,b)\neq(0,0)$ is :

(A) below the x - axis at a distance of $\frac{3}{2}$ from it

(B) below the x - axis at a distance of $\frac{2}{3}$ from it

(C) above the x - axis at a distance of $\frac{3}{2}$ from it

(D) above the x - axis at a distance of $\frac{2}{3}$ from it

Open solution

Q157. AIEEE 2005

If a vertex of a triangle is $(1,1)$ and the mid points of two sides through this vertex are $(-1,2)$ and $(3,2)$ then the centroid of the triangle is :

(A) $(-1,\frac{7}{3})$ (B) $(\frac{-1}{3},\frac{7}{3})$

(C) $(1,\frac{7}{3})$ (D) $(\frac{1}{3},\frac{7}{3})$

Open solution

Q158. AIEEE 2005

If non zero numbers a,b,c are in $H.P.$, then the straight line $\frac{x}{a}+\frac{y}{b}+\frac{1}{c}=0$ always passes through a fixed point. That point is :

(A) $(-1,2)$ (B) $(-1,-2)$

(C) $(1,-2)$ (D) $(1,-\frac{1}{2})$

Open solution

Q159. AIEEE 2004

Let $A\,(2,-3)$ and $B\,(-2,1)$ be vertices of a triangle ABC. If the centroid of this triangle moves on the line $2x+3y=1$, then the locus of the vertex C is the line :

(A) $3x-2y=3$ (B) $2x-3y=7$ (C) $3x+2y=5$ (D) $2x+3y=9$

Open solution

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Q160. AIEEE 2004

The equation of the straight line passing through the point $(4, 3)$ and making intercepts on the co-ordinate axes whose sum is -1 is :

(A) $\frac{x}{2} - \frac{y}{3} = 1$ and $\frac{x}{-2} + \frac{y}{1} = 1$

(B) $\frac{x}{2} - \frac{y}{3} = -1$ and $\frac{x}{-2} + \frac{y}{1} = -1$

(C) $\frac{x}{2} + \frac{y}{3} = 1$ and $\frac{x}{2} + \frac{y}{1} = 1$

(D) $\frac{x}{2} + \frac{y}{3} = -1$ and $\frac{x}{-2} + \frac{y}{1} = -1$

Open solution

Q161. AIEEE 2003

A square of side a lies above the x-axis and has one vertex at the origin. The side passing through the origin makes an angle $\alpha\left(0 < \alpha < \frac{\pi}{4}\right)$ with the positive direction of x-axis. The equation of its diagonal not passing through the origin is :

(A) $y(\cos\alpha + \sin\alpha) + x(\cos\alpha - \sin\alpha) = a$

(B) $y(\cos\alpha - \sin\alpha) - x(\sin\alpha - \cos\alpha) = a$

(C) $y(\cos\alpha + \sin\alpha) + x(\sin\alpha - \cos\alpha) = a$

(D) $y(\cos\alpha + \sin\alpha) + x(\sin\alpha + \cos\alpha) = a$

Open solution

Q162. AIEEE 2003

Locus of centroid of the triangle whose vertices are $(a\cos t, a\sin t), (b\sin t, -b\cos t)$ and $(1, 0)$, where t is a parameter, is :

(A) $(3x+1)^2 + (3y)^2 = a^2 - b^2$

(B) $(3x-1)^2 + (3y)^2 = a^2 - b^2$

(C) $(3x-1)^2 + (3y)^2 = a^2 + b^2$

(D) $(3x+1)^2 + (3y)^2 = a^2 + b^2$

Open solution

Q163. AIEEE 2003

If x_1, x_2, x_3 and y_1, y_2, y_3 are both in G.P. with the same common ratio, then the points $(x_1, y_1), (x_2, y_2)$ and (x_3, y_3) :

(A) are vertices of a triangle

(B) lie on a straight line

(C) lie on an ellipse

(D) lie on a circle

Open solution

Q164. AIEEE 2003

If the equation of the locus of a point equidistant from the points (a_1, b_1) and (a_2, b_2) is $(a_1 - a_2)x + (b_1 - b_2)y + c = 0$, then the value of $'c'$ is :

(A) $\sqrt{{a_1}^2 + {b_1}^2 - {a_2}^2 - {b_2}^2}$

(B) $\frac{1}{2}\left({a_2}^2 + {b_2}^2 - {a_1}^2 - {b_1}^2\right)$

(C) ${a_1}^2 - {a_2}^2 + {b_1}^2 - {b_2}^2$

(D) $\frac{1}{2}\left({a_1}^2 + {a_2}^2 + {b_1}^2 + {b_2}^2\right)$.

Open solution

Q165. AIEEE 2002

A triangle with vertices $(4, 0), (-1, -1), (3, 5)$ is :

(A) isosceles and right angled

(B) isosceles but not right angled

(C) right angled but not isosceles

(D) neither right angled nor isosceles

Open solution

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Q166. AIEEE 2002

Locus of mid point of the portion between the axes of

$x \cos\alpha + y \sin\alpha = p$ where p is constant is :

(A) $x^2 + y^2 = \frac{4}{p^2}$

(B) $x^2 + y^2 = 4p^2$

(C) $\frac{1}{x^2} + \frac{1}{y^2} = \frac{2}{p^2}$

(D) $\frac{1}{x^2} + \frac{1}{y^2} = \frac{4}{p^2}$

Open solution

Numerical 23 questions

Q167. JEE Main 2026 (Online) 5th April Evening Shift

From the point $(-1, -1)$, two rays are sent making angles of 45° with the line $x + y = 0$. These rays get reflected from the mirror $x + 2y = 1$. If the equations of the reflected rays are $ax + by = 9$ and $cx + dy = 7, a, b, c, d \in \mathbf{Z}$, then the value of $ad + bc$ is ____ .

Open solution

Q168. JEE Main 2026 (Online) 4th April Evening Shift

Let A, B be points on the two half-lines $x - \sqrt{3}|y| = \alpha, \alpha > 0$ at a distance of α from their point of intersection P. The line segment AB meets the angle bisector of the given half-lines at the point Q. If $PQ = \frac{9}{2}$ and R is the radius of the circumcircle of $\triangle \mathrm{PAB}$, then $\frac{\alpha^2}{R}$ is equal to ____

Open solution

Q169. JEE Main 2025 (Online) 22nd January Evening Shift

Let the distance between two parallel lines be 5 units and a point P lie between the lines at a unit distance from one of them. An equilateral triangle PQR is formed such that Q lies on one of the parallel lines, while R lies on the other. Then $(QR)^2$ is equal to _____.

Open solution

Q170. JEE Main 2024 (Online) 8th April Evening Shift

Let a ray of light passing through the point $(3, 10)$ reflects on the line $2x + y = 6$ and the reflected ray passes through the point $(7, 2)$. If the equation of the incident ray is $ax + by + 1 = 0$, then $a^2 + b^2 + 3ab$ is equal to _____.

Open solution

Q171. JEE Main 2024 (Online) 8th April Morning Shift

If the orthocentre of the triangle formed by the lines $2x + 3y - 1 = 0, x + 2y - 1 = 0$ and $ax + by - 1 = 0$, is the centroid of another triangle, whose circumcentre and orthocentre respectively are $(3, 4)$ and $(-6, -8)$, then the value of $|a - b|$ is _____.

Open solution

Q172. JEE Main 2024 (Online) 1st February Evening Shift

Let ABC be an isosceles triangle in which A is at $(-1, 0), \angle A = \frac{2\pi}{3}, AB = AC$ and B is on the positve x-axis. If $\mathrm{BC} = 4\sqrt{3}$ and the line BC intersects the line $y = x + 3$ at (α, β), then $\frac{\beta^4}{\alpha^2}$ is _____.

Open solution

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Q173. JEE Main 2024 (Online) 1st February Evening Shift

The lines $L_1, L_2, \ldots, L_{20}$ are distinct. For $n = 1, 2, 3, \ldots, 10$ all the lines L_{2n-1} are parallel to each other and all the lines L_{2n} pass through a given point P. The maximum number of points of intersection of pairs of lines from the set $\{L_1, L_2, \ldots, L_{20}\}$ is equal to _____.

Open solution

Q174. JEE Main 2024 (Online) 31st January Evening Shift

Let $A(-2,-1), B(1,0), C(\alpha,\beta)$ and $D(\gamma,\delta)$ be the vertices of a parallelogram $ABCD$. If the point C lies on $2x - y = 5$ and the point D lies on $3x - 2y = 6$, then the value of $|\alpha + \beta + \gamma + \delta|$ is equal to _____.

Open solution

Q175. JEE Main 2024 (Online) 27th January Evening Shift

If the sum of squares of all real values of α, for which the lines $2x - y + 3 = 0, 6x + 3y + 1 = 0$ and $\alpha x + 2y - 2 = 0$ do not form a triangle is p, then the greatest integer less than or equal to p is _____.

Open solution

Q176. JEE Main 2023 (Online) 10th April Evening Shift

Let the equations of two adjacent sides of a parallelogram ABCD be $2x - 3y = -23$ and $5x + 4y = 23$. If the equation of its one diagonal AC is $3x + 7y = 23$ and the distance of A from the other diagonal is d, then $50\,d^2$ is equal to _____.

Open solution

Q177. JEE Main 2023 (Online) 24th January Evening Shift

The equations of the sides AB, BC and CA of a triangle ABC are : $2x + y = 0, x + py = 21a, (a \neq 0)$ and $x - y = 3$ respectively. Let P(2, a) be the centroid of ΔABC. Then $(BC)^2$ is equal to _____.

Open solution

Q178. JEE Main 2022 (Online) 30th July Evening Shift

Let $A(0, a+2), B(0, a), C(-2, 0)$ and $D(2, 0)$ be four points and the lines AD and BC intersect at $P(x, y)$. If the locus of P is the curve $f(x, y) = 0$ and the tangent at the point $(4, \gamma)$ on this curve is $\frac{x}{\alpha} + \frac{y}{\beta} = 1$, then $\alpha(\beta - \gamma)$ is equal to _____.

Open solution

Q179. JEE Main 2022 (Online) 26th July Morning Shift

The equations of the sides AB, BC and CA of a triangle ABC are $2x + y = 0, x + py = 15a$ and $x - y = 3$ respectively. If its orthocentre is $(2, a), -\frac{1}{2} < a < 2$, then p is equal to _____.

Open solution

Q180. JEE Main 2022 (Online) 28th June Morning Shift

A ray of light passing through the point P(2, 3) reflects on the x-axis at point A and the reflected ray passes through the point Q(5, 4). Let R be the point that divides the line segment AQ internally into the ratio 2 : 1. Let the co-ordinates of the foot of the perpendicular M from R on the bisector of the angle PAQ be (α, β). Then, the value of $7\alpha + 3\beta$ is equal to _____.

Open solution

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Q181. JEE Main 2022 (Online) 24th June Morning Shift

Let $A\left(\frac{3}{\sqrt{a}}, \sqrt{a}\right)$, $a > 0$, be a fixed point in the xy-plane. The image of A in y-axis be B and the image of B in x-axis be C. If $D(3\cos\theta, a\sin\theta)$ is a point in the fourth quadrant such that the maximum area of ΔACD is 12 square units, then a is equal to _____.

Open solution

Q182. JEE Main 2021 (Online) 1st September Evening Shift

Let the points of intersections of the lines x $-$ y + 1 = 0, x $-$ 2y + 3 = 0 and 2x $-$ 5y + 11 = 0 are the mid points of the sides of a triangle ΔABC. Then, the area of the ΔABC is _____.

Open solution

Q183. JEE Main 2021 (Online) 1st September Evening Shift

A man starts walking from the point P($-$3, 4), touches the x-axis at R, and then turns to reach at the point Q(0, 2). The man is walking at a constant speed. If the man reaches the point Q in the minimum time, then $50\left((PR)^2 + (RQ)^2\right)$ is equal to _____.

Open solution

Q184. JEE Main 2021 (Online) 20th July Evening Shift

Consider a triangle having vertices A($-$2, 3), B(1, 9) and C(3, 8). If a line L passing through the circum-centre of triangle ABC, bisects line BC, and intersects y-axis at point $\left(0, \frac{\alpha}{2}\right)$, then the value of real number α is _____.

Open solution

Q185. JEE Main 2021 (Online) 18th March Morning Shift

A square ABCD has all its vertices on the curve $x^2y^2 = 1$. The midpoints of its sides also lie on the same curve. Then, the square of area of ABCD is _____.

Open solution

Q186. JEE Main 2021 (Online) 17th March Evening Shift

Let $\tan\alpha$, $\tan\beta$ and $\tan\gamma$; $\alpha, \beta, \gamma \neq \frac{(2n-1)\pi}{2}$, n$\in$N be the slopes of three line segments OA, OB and OC, respectively, where O is origin. If circumcentre of ΔABC coincides with origin and its orthocentre lies on y-axis, then the value of $\left(\frac{\cos 3\alpha+\cos 3\beta+\cos 3\gamma}{\cos\alpha\cos\beta\cos\gamma}\right)^2$ is equal to _____.

Open solution

Q187. JEE Main 2021 (Online) 17th March Morning Shift

The maximum value of z in the following equation z = 6xy + y^2, where 3x + 4y $\leq$ 100 and 4x + 3y $\leq$ 75 for x $\geq$ 0 and y $\geq$ 0 is _____.

Open solution

Q188. JEE Main 2020 (Online) 5th September Morning Slot

If the line, 2x - y + 3 = 0 is at a distance $\frac{1}{\sqrt{5}}$ and $\frac{2}{\sqrt{5}}$ from the lines 4x - 2y + α = 0 and 6x - 3y + β = 0, respectively, then the sum of all possible values of α and β is :

Open solution

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Q189. JEE Main 2020 (Online) 7th January Morning Slot

Let A(1, 0), B(6, 2) and C $\left(\frac{3}{2}, 6\right)$ be the vertices of a triangle ABC. If P is a Point inside the triangle ABC such that the triangles APC, APB and BPC have equal areas, then the length of the line segment PQ, where Q is the point $\left(-\frac{7}{6}, -\frac{1}{3}\right)$, is _____.

Open solution

Out of syllabus 17 questions

These questions are on topics no longer in the current syllabus. They are kept for practice, after everything else.

Q190. JEE Main 2023 (Online) 8th April Evening Shift · MCQ (Single Correct Answer)

Let $\mathrm{A}(0,1), \mathrm{B}(1,1)$ and $\mathrm{C}(1,0)$ be the mid-points of the sides of a triangle with incentre at the point D. If the focus of the parabola $y^2 = 4\mathrm{ax}$ passing through D is $(\alpha + \beta\sqrt{2}, 0)$, where α and β are rational numbers, then $\frac{\alpha}{\beta^2}$ is equal to :

(A) $\frac{9}{2}$ (B) 12 (C) 6 (D) 8

Open solution

Q191. JEE Main 2023 (Online) 1st February Morning Shift · MCQ (Single Correct Answer)

The combined equation of the two lines $ax + by + c = 0$ and $a'x + b'y + c' = 0$ can be written as

$(ax + by + c)(a'x + b'y + c') = 0.$

The equation of the angle bisectors of the lines represented by the equation $2x^2 + xy - 3y^2 = 0$ is :

(A) $3x^2 + xy - 2y^2 = 0$ (B) $x^2 - y^2 - 10xy = 0$

(C) $x^2 - y^2 + 10xy = 0$ (D) $3x^2 + 5xy + 2y^2 = 0$

Open solution

Q192. JEE Main 2023 (Online) 30th January Morning Shift · MCQ (Single Correct Answer)

A straight line cuts off the intercepts $\mathrm{OA} = \mathrm{a}$ and $\mathrm{OB} = \mathrm{b}$ on the positive directions of x-axis and y axis respectively. If the perpendicular from origin O to this line makes an angle of $\frac{\pi}{6}$ with positive direction of y-axis and the area of $\triangle\mathrm{OAB}$ is $\frac{98}{3}\sqrt{3}$, then $\mathrm{a}^2 - \mathrm{b}^2$ is equal to :

(A) $\frac{392}{3}$ (B) 98 (C) 196 (D) $\frac{196}{3}$

Open solution

Q193. JEE Main 2021 (Online) 27th August Evening Shift · MCQ (Single Correct Answer)

Let A(a, 0), B(b, 2b + 1) and C(0, b), $b \neq 0$, $|b| \neq 1$, be points such that the area of triangle ABC is 1 sq. unit, then the sum of all possible values of a is :

(A) $\frac{-2b}{b+1}$ (B) $\frac{2b}{b+1}$ (C) $\frac{2b^2}{b+1}$ (D) $\frac{-2b^2}{b+1}$

Open solution

Q194. JEE Main 2021 (Online) 26th August Morning Shift · MCQ (Single Correct Answer)

Let ABC be a triangle with A$(-3, 1)$ and $\angle$ACB = θ, $0 < \theta < \frac{\pi}{2}$. If the equation of the median through B is $2x + y - 3 = 0$ and the equation of angle bisector of C is $7x - 4y - 1 = 0$, then $\tan\theta$ is equal to :

(A) $\frac{1}{2}$ (B) $\frac{3}{4}$ (C) $\frac{4}{3}$ (D) 2

Open solution

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Q195. JEE Main 2021 (Online) 25th July Evening Shift · MCQ (Single Correct Answer)

Let the equation of the pair of lines, y = px and y = qx, can be written as $(y - px)(y - qx) = 0$. Then the equation of the pair of the angle bisectors of the lines $x^2 - 4xy - 5y^2 = 0$ is :

(A) $x^2 - 3xy + y^2 = 0$

(B) $x^2 + 4xy - y^2 = 0$

(C) $x^2 + 3xy - y^2 = 0$

(D) $x^2 - 3xy - y^2 = 0$

Open solution

Q196. JEE Main 2019 (Online) 12th April Evening Slot · MCQ (Single Correct Answer)

A triangle has a vertex at (1, 2) and the mid points of the two sides through it are (−1, 1) and (2, 3). Then the centroid of this triangle is :

(A) $\left(\frac{1}{3}, 2\right)$

(B) $\left(\frac{1}{3}, \frac{5}{3}\right)$

(C) $\left(1, \frac{7}{3}\right)$

(D) $\left(\frac{1}{3}, 1\right)$

Open solution

Q197. JEE Main 2019 (Online) 9th January Morning Slot · MCQ (Single Correct Answer)

Consider the set of all lines px + qy + r = 0 such that 3p + 2q + 4r = 0. Which one of the following statements is true?

(A) The lines are not concurrent

(B) The lines are concurrent at the point $\left(\frac{3}{4}, \frac{1}{2}\right)$

(C) The lines are all parallel

(D) Each line passes through the origin

Open solution

Q198. AIEEE 2011 · MCQ (Single Correct Answer)

The lines $L_1 : y - x = 0$ and $L_2 : 2x + y = 0$ intersect the line $L_3 : y + 2 = 0$ at P and Q respectively. The bisector of the acute angle between L_1 and L_2 intersects L_3 at R.

Statement-1: The ratio $PR : RQ$ equals $2\sqrt{2} : \sqrt{5}$

Statement-2: In any triangle, bisector of an angle divide the triangle into two similar triangles.

(A) Statement-1 is true, Statement-2 is true; Statement-2 is not a correct explanation for Statement-1.

(B) Statement-1 is true, Statement-2 is false.

(C) Statement-1 is false, Statement-2 is true.

(D) Statement-1 is true, Statement-2 is true; Statement-2 is a correct explanation for Statement-1.

Open solution

Q199. AIEEE 2007 · MCQ (Single Correct Answer)

Let $P = (-1, 0)$, $Q = (0, 0)$ and $R = \left(3, 3\sqrt{3}\right)$ be three point. The equation of the bisector of the angle PQR is :

(A) $\frac{\sqrt{3}}{2}x + y = 0$

(B) $x + \sqrt{3y} = 0$

(C) $\sqrt{3}x + y = 0$

(D) $x + \frac{\sqrt{3}}{2}y = 0$

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Q200. AIEEE 2007 · MCQ (Single Correct Answer)

If one of the lines of $my^2 + (1 - m^2)xy - mx^2 = 0$ is a bisector of angle between the lines $xy = 0$, then m is :

(A) 1 (B) 2 (C) $-1/2$ (D) -2

Open solution

Q201. AIEEE 2004 · MCQ (Single Correct Answer)

If the sum of the slopes of the lines given by $x^2 - 2cxy - 7y^2 = 0$ is four times their product, then c has the value :

(A) -2 (B) -1 (C) 2 (D) 1

Open solution

Q202. AIEEE 2004 · MCQ (Single Correct Answer)

If one of the lines given by $6x^2 - xy + 4cy^2 = 0$ is $3x + 4y = 0$, then c equals :

(A) -3 (B) -1 (C) 3 (D) 1

Open solution

Q203. AIEEE 2003 · MCQ (Single Correct Answer)

If the pair of straight lines $x^2 - 2pxy - y^2 = 0$ and $x^2 - 2qxy - y^2 = 0$ be such that each pair bisects the angle between the other pair, then :

(A) $pq = -1$ (B) $p = q$ (C) $p = -q$ (D) $pq = 1$.

Open solution

Q204. AIEEE 2002 · MCQ (Single Correct Answer)

If the pair of lines

$$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$$

intersect on the y-axis then :

(A) $2fgh = bg^2 + ch^2$ (B) $bg^2 \neq ch^2$ (C) $abc = 2fgh$ (D) none of these

Open solution

Q205. AIEEE 2002 · MCQ (Single Correct Answer)

The pair of lines represented by $3ax^2 + 5xy + (a^2 - 2)y^2 = 0$

are perpendicular to each other for :

(A) two values of a (B) $\forall\, a$

(C) for one value of a (D) for no values of a

Open solution

Q206. JEE Main 2023 (Online) 11th April Evening Shift · Numerical

If the line $l_1 : 3y - 2x = 3$ is the angular bisector of the lines $l_2 : x - y + 1 = 0$ and $l_3 : \alpha x + \beta y + 17 = 0$, then $\alpha^2 + \beta^2 - \alpha - \beta$ is equal to _____.

Open solution

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Answer Key

MCQ (Single Correct Answer)

1. **B**	2. **D**	3. **A**	4. **C**	5. **C**
6. **B**	7. **A**	8. **B**	9. **B**	10. **C**
11. **C**	12. **A**	13. **A**	14. **B**	15. **A**
16. **D**	17. **C**	18. **B**	19. **B**	20. **D**
21. **C**	22. **B**	23. **D**	24. **D**	25. **B**
26. **A**	27. **D**	28. **A**	29. **C**	30. **C**
31. **B**	32. **A**	33. **C**	34. **A**	35. **D**
36. **A**	37. **B**	38. **B**	39. **C**	40. **A**
41. **B**	42. **D**	43. **D**	44. **C**	45. **A**
46. **B**	47. **A**	48. **C**	49. **B**	50. **B**
51. **B**	52. **C**	53. **A**	54. **D**	55. **D**
56. **B**	57. **B**	58. **B**	59. **D**	60. **C**
61. **B**	62. **D**	63. **C**	64. **B**	65. **C**
66. **C**	67. **B**	68. **C**	69. **D**	70. **C**
71. **C**	72. **A**	73. **C**	74. **B**	75. **B**
76. **B**	77. **B**	78. **A**	79. **D**	80. **A**
81. **D**	82. **D**	83. **D**	84. **A**	85. **A**
86. **B**	87. **A**	88. **D**	89. **D**	90. **D**
91. **D**	92. **D**	93. **B**	94. **B**	95. **D**
96. **D**	97. **A**	98. **A**	99. **A**	100. **B**
101. **D**	102. **B**	103. **D**	104. **C**	105. **B**
106. **B**	107. **D**	108. **D**	109. **A**	110. **D**
111. **D**	112. **B**	113. **C**	114. **A**	115. **C**
116. **C**	117. **C**	118. **A**	119. **B**	120. **D**
121. **A**	122. **B**	123. **D**	124. **B**	125. **B**
126. **C**	127. **B**	128. **C**	129. **D**	130. **D**
131. **A**	132. **B**	133. **D**	134. **B**	135. **A**
136. **D**	137. **D**	138. **A**	139. **B**	140. **B**
141. **C**	142. **D**	143. **C**	144. **D**	145. **B**
146. **C**	147. **A**	148. **C**	149. **C**	150. **D**

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151. **A**	152. **D**	153. **A**	154. **C**	155. **C**
156. **A**	157. **C**	158. **C**	159. **D**	160. **A**
161. **A**	162. **C**	163. **B**	164. **B**	165. **A**
166. **D**				

Numerical

167. **7**	168. **9**	169. **28**	170. **1**	171. **16**
172. **36**	173. **101**	174. **32**	175. **32**	176. **529**
177. **122**	178. **4**	179. **3**	180. **31**	181. **8**
182. **6**	183. **1250**	184. **9**	185. **80**	186. **144**
187. **904**	188. **30**	189. **5**		

Out of syllabus

190. **D**	191. **B**	192. **A**	193. **D**	194. **C**
195. **C**	196. **A**	197. **B**	198. **B**	199. **C**
200. **A**	201. **C**	202. **A**	203. **A**	204. **A**
205. **A**	206. **348**			

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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